

INDUSTRIAL UTILITY & GENERAL SERVICE PIPING - PSL 1 QUALITY PLAN

Pipe Description: 16" (406.4 mm) x 9.53 mm, Grade B PSL 1, SMLS (Seamless Pipe) | **Governing Standard:** API Specification 5L 47th Edition PSL 1

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C <= 0.28%, P <= 0.030%, S <= 0.030% (PSL 1 limits)	API 5L Tablo 4	H/W
2	Product Chemical Analysis	Two analyses per heat of steel	Pipe body coupon	ISO 14284	C <= 0.28%, P <= 0.030%, S <= 0.030%	API 5L Tablo 4	W
3	Pipe Body Tensile Test	Once per test unit (lot <= 400 pipes)	Longitudinal body strap	ASTM A370	Rt0.5 >= 245 MPa (35.5 ksi), Rm >= 415 MPa (60.2 ksi), Af >= 27.5%	API 5L Tablo 6	H/W
4	Mill Hydrostatic Pressure Test	Each pipe (100% of all pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 70 bar, Duration: MINIMUM 5 SECONDS (D < 457 mm)	API 5L 10.2.6.2	H/W
5	Dimensional & Wall Thickness Verification	Sample per lot and 100% pipe ends	Pipe body & ends	API 5L 10.2.8	Diameter: +-0.75%; Wall thickness: -12.5% / +15.0% (PSL 1 Table 11)	API 5L Tablo 10/11	W
6	Visual Surface & Marking Inspection	100% of all pipes	Full pipe surface	API 5L 10.2.7	Visual surface free from defects; API 5L Grade B PSL 1 paint stencil marking	API 5L Madde 11	W